TUGAS PERTEMUAN 10 DASAR PEMROGRAMAN

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Kelas: TI – 1C

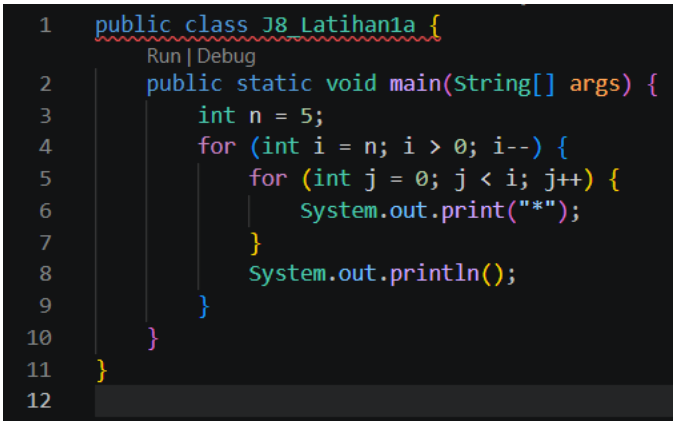
NIM: 264107020073

Latihan Soal 1a

Pseudocode:

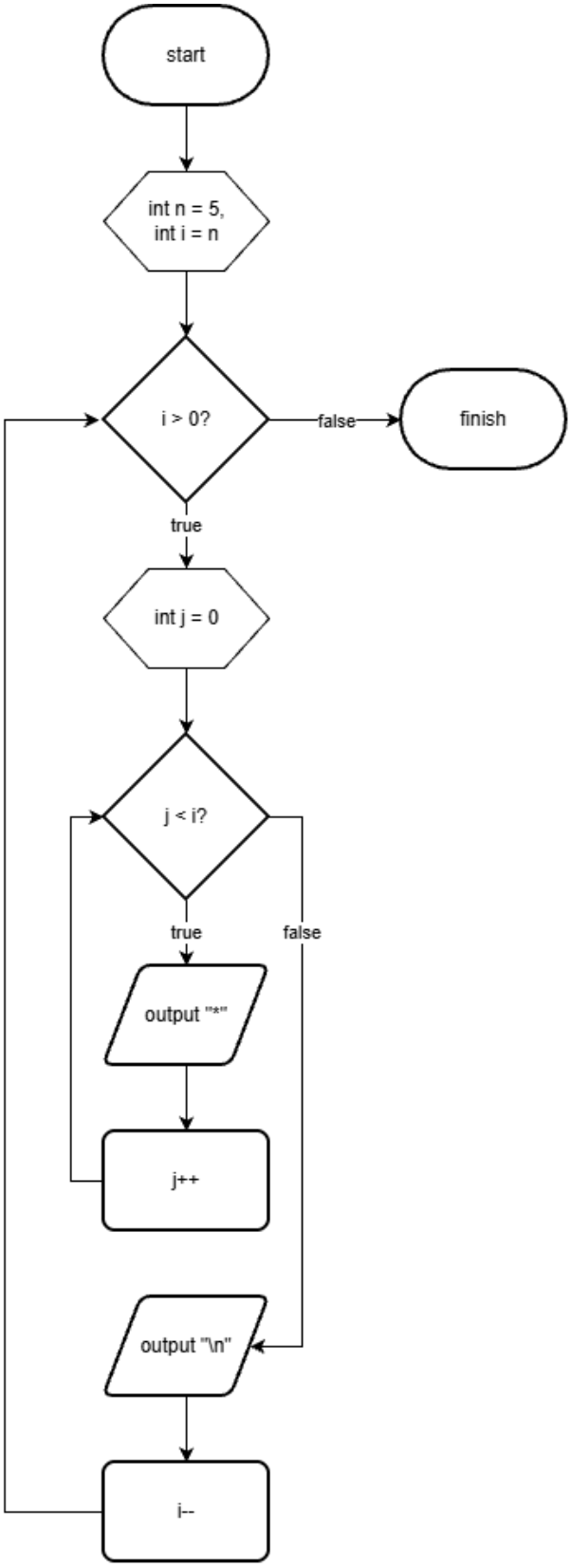
PROGRAM J8_Latihan1a
DEKLARASI n: int = 5
ALGORITMA **FOR** i = n **TO** i > 0 **STEP** i-- **DO**   **FOR** j = 0 **TO** j < i **STEP** j++ **DO**   **print** “*”   **END FOR**   **print** “\n” **END FOR**

Program:



```
1   public class J8_Latihan1a {
        Run | Debug
2       public static void main(String[] args) {
3           int n = 5;
4           for (int i = n; i > 0; i--) {
5               for (int j = 0; j < i; j++) {
6                   System.out.print("*");
7               }
8               System.out.println();
9           }
10      }
11  }
12
```

Flowchart:


start
int n = 5,
int i = n
i > 0?
false
finish
true
int j = 0
j < i?
true
false
output "*"
j++
output "\n"
i--

Latihan Soal 1b

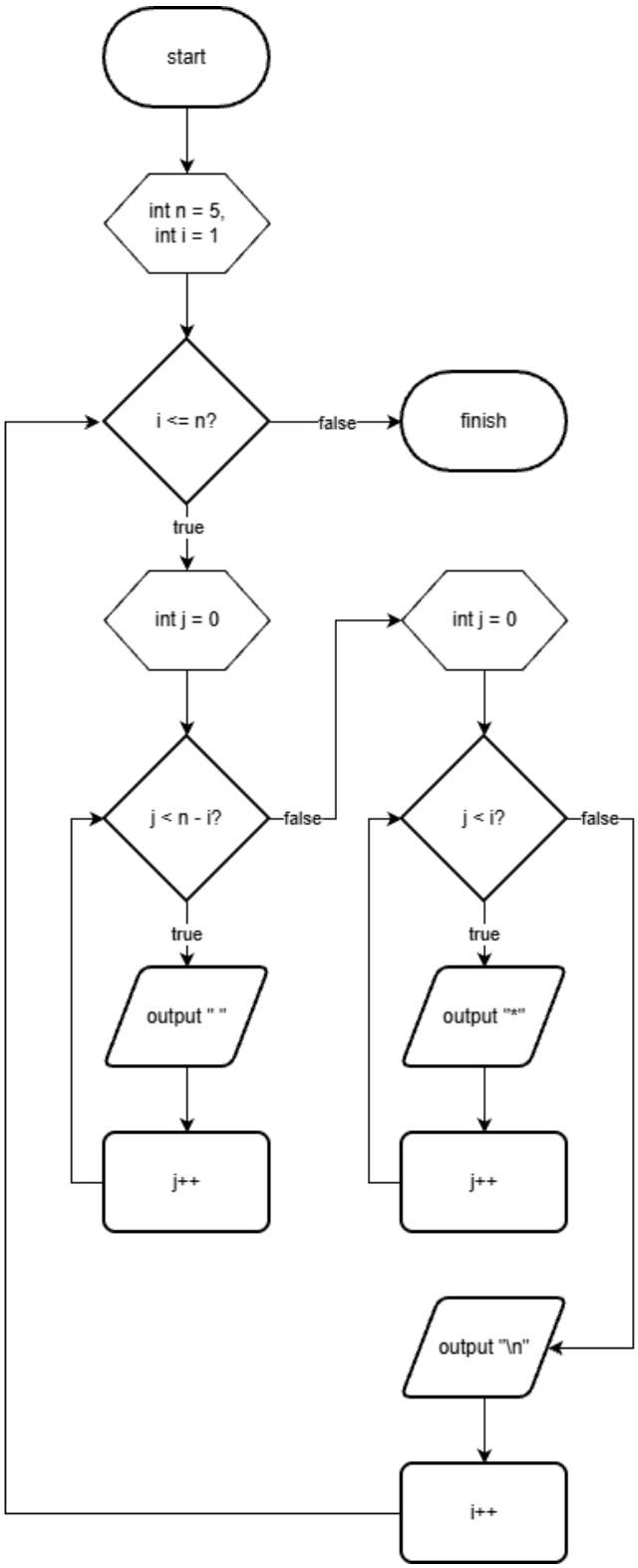
Pseudocode:

PROGRAM J8_Latihan1b
DEKLARASI n : int = 5
ALGORITMA **FOR** i = 1 **TO** i <= n **STEP** i++ **DO**   **FOR** j = 0 **TO** j < n - i **STEP** j++ **DO**   **print** “ ”   **END FOR**   **FOR** j = 0 **TO** j < i **STEP** j++ **DO**   **print** “*”   **END FOR**   **print** “\n” **END FOR**

Program:

```java
public class J8_Latihan1b {
    Run | Debug
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 0; j < n - i; j++) {
                System.out.print(" ");
            }
            for (int j = 0; j < i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Flowchart:


start
int n = 5,
int i = 1
i <= n?
false
finish
true
int j = 0
int j = 0
j < n - i?
false
j < i?
false
true
true
output " "
output "*"
j++
j++
output "\n"
i++

Latihan Soal 2

Pseudocode:

```
PROGRAM
J8_Latihan2
```

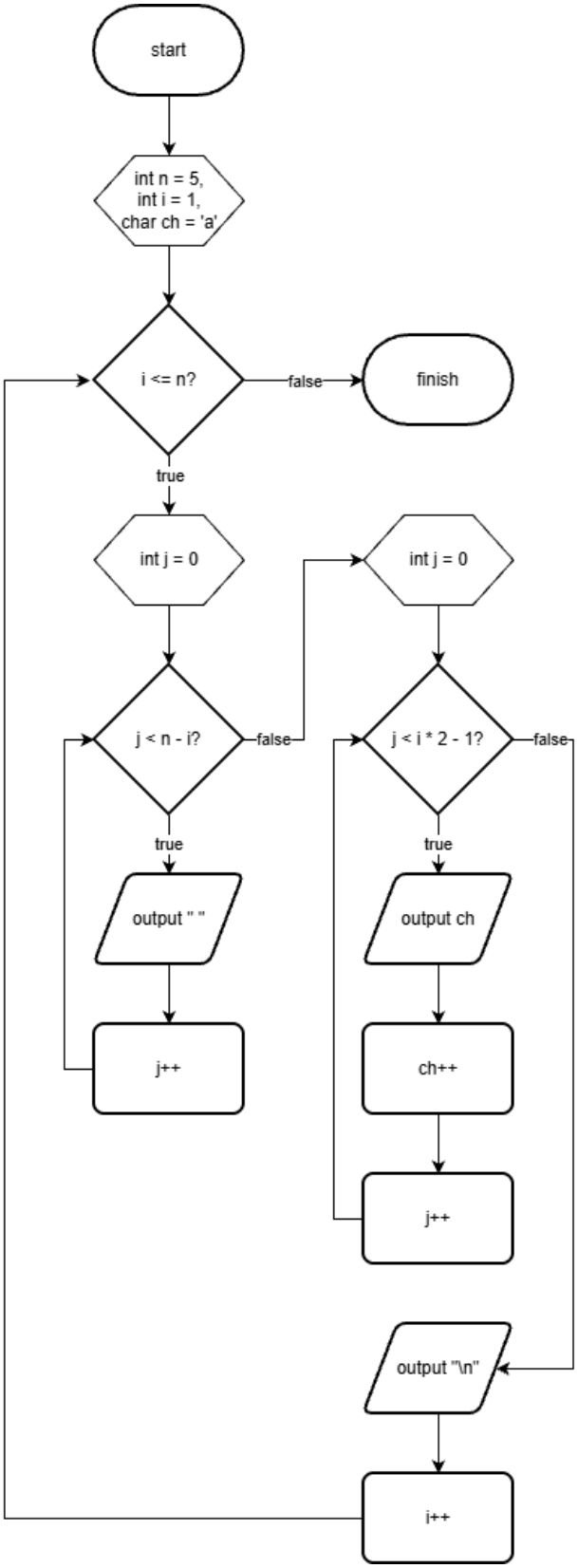
```
DEKLARASI
n : int = 5
ch : char = ‘a’
```

```
ALGORITMA
FOR i = 1 TO i <= n STEP i++ DO
      FOR j = 0 TO j < n - i STEP j++ DO
            print “ ”
      END FOR
      FOR j = 0 TO j < i * 2 - 1 STEP j++ DO
            print ch
            ch++
      END FOR
      print “\n”
END FOR
```

Program:

```java
public class J8_Latihan2 {
    Run | Debug
    public static void main(String[] args) {
        int n = 5;
        char ch = 'a';
        for (int i = 1; i <= n; i++) {
            for (int j = 0; j < n - i; j++) {
                System.out.print(" ");
            }
            for (int j = 0; j < i * 2 - 1; j++) {
                System.out.print(ch);
                ch++;
            }
            System.out.println();
        }
    }
}
```

Flowchart:


start
int n = 5,
int i = 1,
char ch = 'a'
i <= n?
false
finish
true
int j = 0
j < n - i?
false
true
output " "
j++
int j = 0
j < i * 2 - 1?
false
true
output ch
ch++
j++
output "\n"
i++

Latihan Soal 3

Pseudocode:

```
PROGRAM
J8_Latihan3

DEKLARASI
n : int = 5
jumlah : int

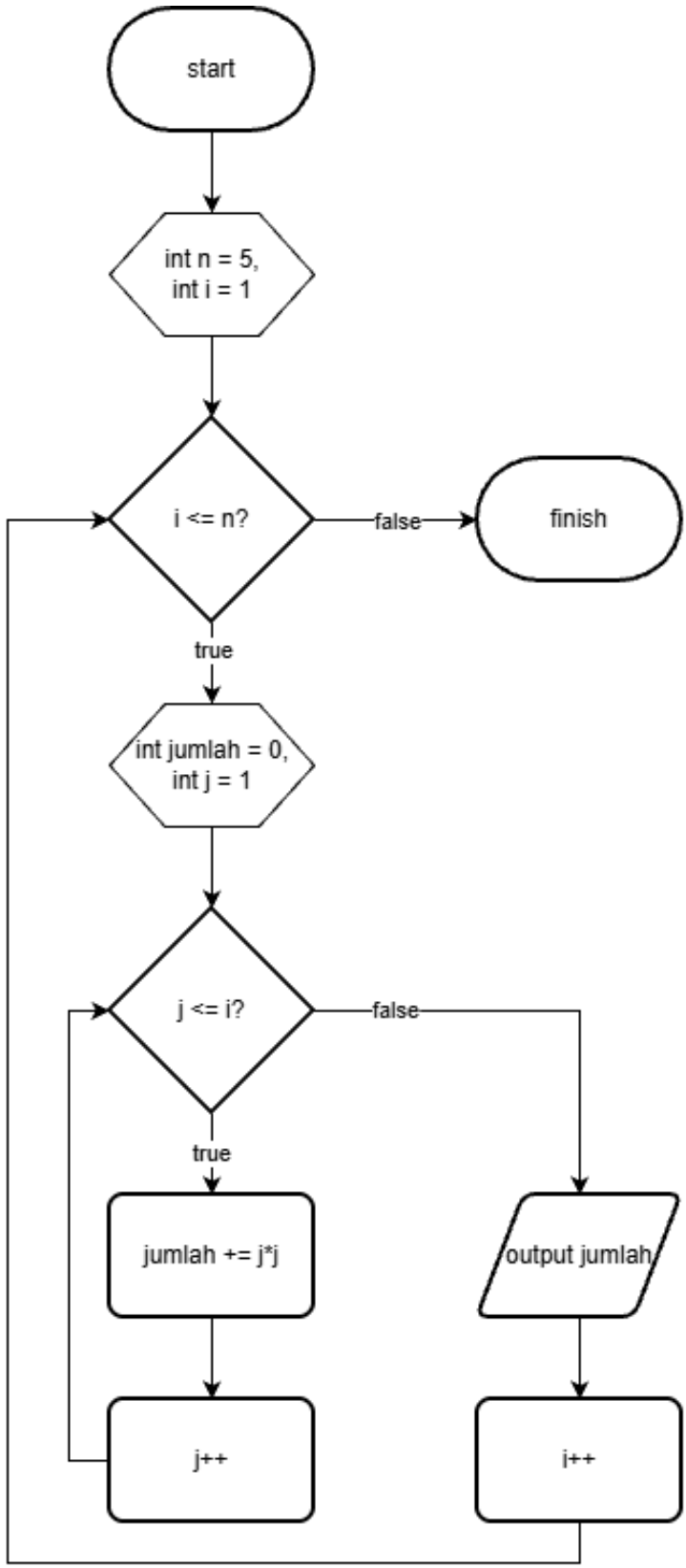
ALGORITMA
FOR i = 1 TO i <= n STEP i++ DO
      print "n = " + i + " -> jumlah kuadrat = "
      jumlah = 0
      FOR j = 1 TO j <= i STEP j++ DO
            print " "
            jumlah += j*j
            IF i == 1 THEN
                  Break
            END IF
            print j*j
            IF j < i THEN
                  print " + "
            ELSE
                  print " = "
            END IF
      END FOR
      print "\n"
END FOR
```

Program:

```java
public class J8_Latihan3 {
    Run | Debug
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            System.out.print("n = " + i + " -> jumlah kuadrat = ");
            int jumlah = 0;
            for (int j = 1; j <= i; j++) {
                jumlah += j*j;
                if (i == 1) {
                    break;
                }
                System.out.print(j*j);
                if (j < i) {
                    System.out.print(" + ");
                } else {
                    System.out.print(" = ");
                }
            }
            System.out.println(jumlah);
        }
    }
}

```

Flowchart:


start
int n = 5,
int i = 1
i <= n?
false
finish
true
int jumlah = 0,
int j = 1
j <= i?
false
true
jumlah += j*j
output jumlah
j++
i++

Latihan Soal 4

Pseudocode:

```
PROGRAM
J8_Latihan4
```

```
DEKLARASI
totalKeseluruhan : int
banyakProduk : int
jumlahHari : int
totalPerProduk : int
rata : double
jumlah : int
```

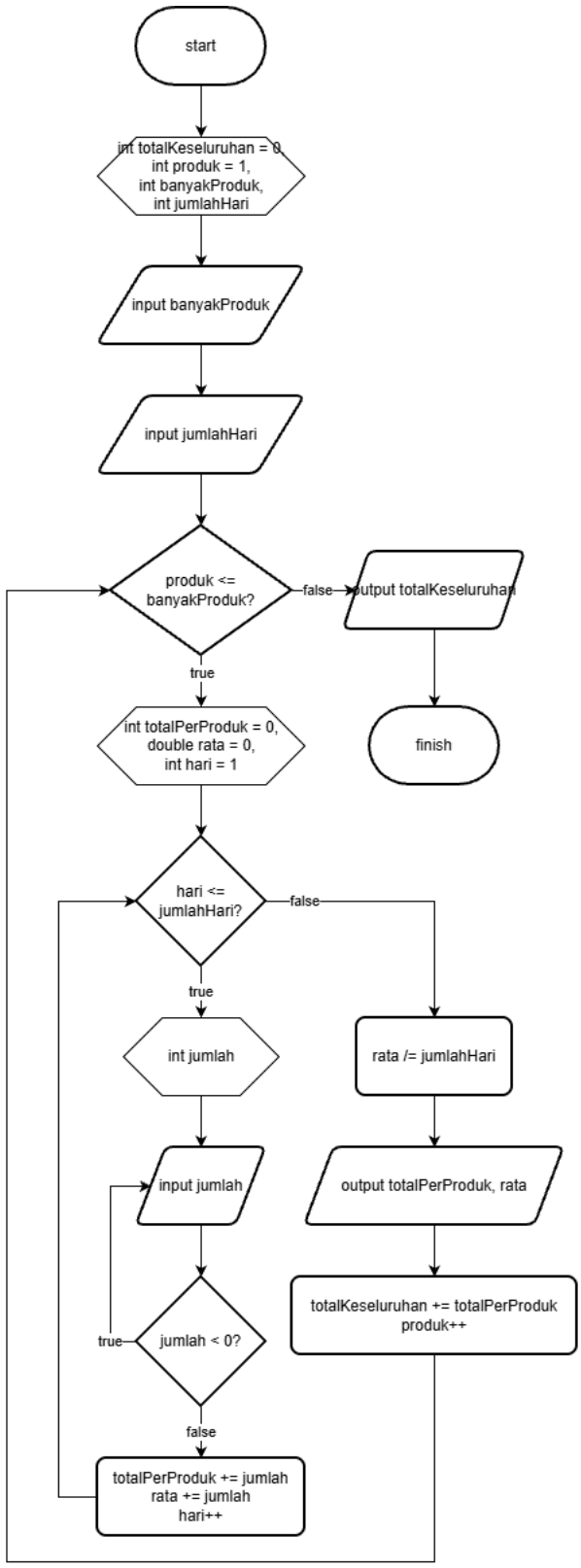
```
ALGORITMA
totalKeseluruhan = 0
input banyakProduk
input jumlahHari
produk = 1
WHILE produk <= banyakProduk DO
      totalPerProduk = 0
      rata = 0
      FOR hari = 1 TO hari <= jumlahHari STEP hari++ DO
            DO
                  input jumlah
            WHILE jumlah < 0
            totalPerProduk += jumlah
            rata += jumlah
      END FOR
      rata /= jumlahHari
      print totalPerProduk, rata
      print "\n"
      totalKeseluruhan += totalPerProduk
      produk++
END FOR
print totalKeseluruhan
```

Program:

```java
import java.util.Scanner;

public class J8_Latihan4 {
    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int totalKeseluruhan = 0;
        System.out.print("Masukkan banyak produk: ");
        int banyakProduk = input.nextInt();
        System.out.print("Masukkan jumlah hari: ");
        int jumlahHari = input.nextInt();
        int produk = 1;
        while (produk <= banyakProduk) {
            int totalPerProduk = 0;
            double rata = 0;
            System.out.println("Produk " + produk + ": ");
            for (int hari = 1; hari <= jumlahHari; hari++) {
                int jumlah;
                do {
                    System.out.print("    Jumlah produk hari ke-" + hari + ": ");
                    jumlah = input.nextInt();
                } while (jumlah < 0);
                totalPerProduk += jumlah;
                rata += jumlah;
            }
            rata /= jumlahHari;
            System.out.println("Total penjualan produk " + produk + ": " + totalPerProduk + ", rata-rata: " + rata);
            totalKeseluruhan += totalPerProduk;
            produk++;
        }
        System.out.println("Total seluruh produk: " + totalKeseluruhan);
        input.close();
    }
}
```

Flowchart:


start
int totalKeseluruhan = 0,
int produk = 1,
int banyakProduk,
int jumlahHari
input banyakProduk
input jumlahHari
produk <=
banyakProduk?
false
output totalKeseluruhan
true
finish
int totalPerProduk = 0,
double rata = 0,
int hari = 1
hari <=
jumlahHari?
false
true
int jumlah
rata /= jumlahHari
input jumlah
output totalPerProduk, rata
totalKeseluruhan += totalPerProduk
produk++
jumlah < 0?
true
false
totalPerProduk += jumlah
rata += jumlah
hari++